

Ch 3: Interdependence and the Gains from Trade

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EC 111 Principles of Microeconomics, Bentley University, Fall 2026

This chapter

- Why do people, and countries, choose to depend on each other?
- How can trade make everyone better off?
- Absolute advantage vs. comparative advantage: what is the difference, and which one matters?
- What determines the price at which two parties trade?

Question

“Trade with China makes most Americans better off because, among other things, they can buy goods that are made more cheaply there.” Agree, disagree, or uncertain?

Interdependence

- Almost everything you consume was produced by people you will never meet.
- Each of them produced something for you and got something in return.
- Ten Principles, no. 5: **trade can make everyone better off.**
- Today: why that is true, even when one side is better at producing everything.

Imagine a week in which you had to grow your own food, make your own clothes, and build your own phone.

A simple world

Two countries, the U.S. and Japan. Two goods, computers and wheat.
One input, labor hours per month.

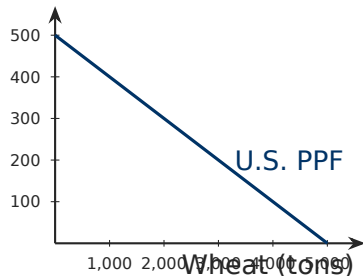
	Labor hours	Hours per computer	Hours per ton of wheat
U.S.	50,000	100	10
Japan	30,000	125	25

Plan: find what each country produces and consumes

1. if it is self-sufficient,
2. if it specializes and trades.

The U.S. production possibilities frontier

Computers

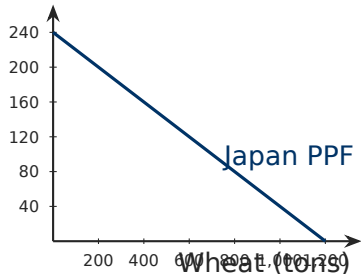


Hours for		Output	
Comp.	Wheat	Comp.	Wheat
50,000	0	500	0
40,000	10,000	400	1,000
25,000	25,000	250	2,500
10,000	40,000	100	4,000
0	50,000	0	5,000

500 computers, or 5,000 tons of wheat, or any point on the line.

Japan's PPF

Computers

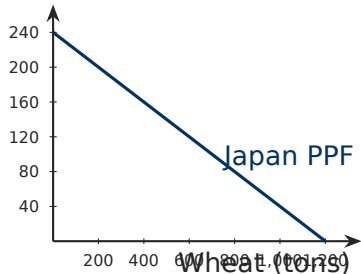


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Japan has 30,000 hours; a computer takes 125 hours, a ton of wheat 25. Find the endpoints of Japan's PPF.

Japan's PPF

Computers



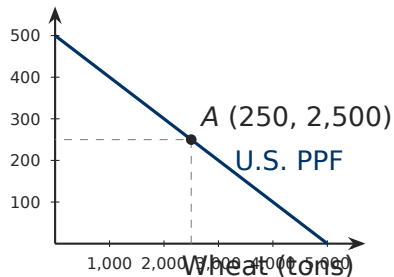
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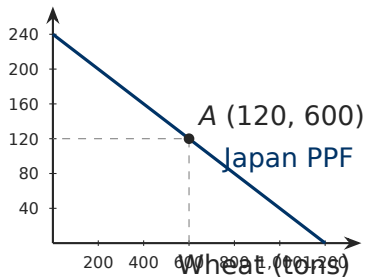
- All computers:
 $30,000 / 125 = 240$
- All wheat: $30,000 / 25 = 1,200$ tons

Without trade: consume what you produce

Computers



Computers



Each country puts half its labor into each good. U.S.: 250 computers and 2,500 tons of wheat. Japan: 120 computers and 600 tons. No trade means the consumption point is **on** the PPF.

Production with trade

Now let each country produce a different bundle from the one it consumes.

Question

The U.S. produces 3,400 tons of wheat. How many computers can it make with the remaining hours?

Japan produces 240 computers. How much wheat can it make with the remaining hours?

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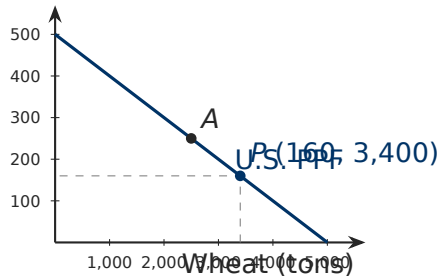
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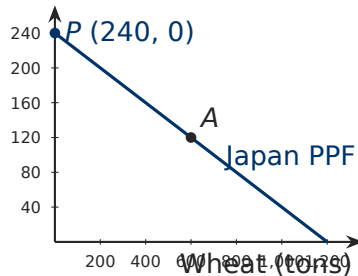
- U.S.: $3,400 \times 10 = 34,000$ h on wheat; 16,000 h left;
 $16,000/100 = 160$ computers.
- Japan: $240 \times 125 = 30,000$ h, all of it; 0 wheat.

Production points

Computers



Computers



The U.S. moved toward wheat; Japan went all the way to computers.
Both P points are still on the PPFs.

Exports and imports

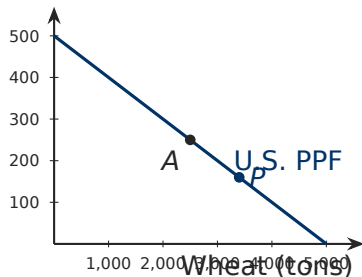
Exports goods produced at home and sold abroad

Imports goods produced abroad and bought at home

Trade proposal: the U.S. sends 700 tons of wheat to Japan; Japan sends 110 computers to the U.S. With this trade, how much of each good does each country end up consuming?

U.S. consumption with trade

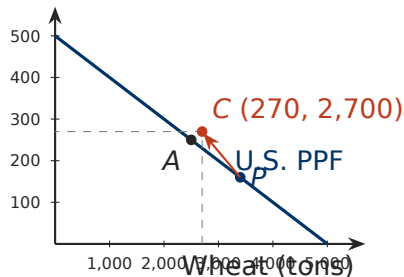
Computers



	Computers	Wheat
Produced	160	3,400
+ Imported	110	0
– Exported	0	700
= Consumed	270	2,700

U.S. consumption with trade

Computers

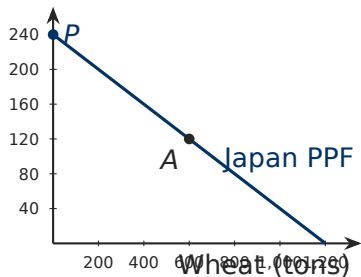


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Point C lies **outside** the U.S. PPF. The U.S. could not produce this bundle on its own.

Japan's consumption with trade

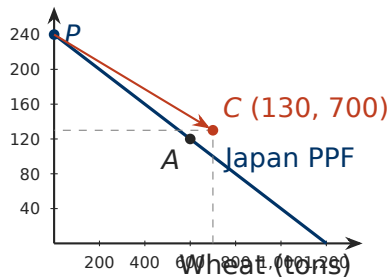
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	Computers	Wheat
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Japan's consumption with trade

Computers



	Computers	Wheat
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Japan produces no wheat, yet consumes 700 tons. Its C is also outside its PPF.

Trade makes both countries better off

United States

	No trade	Trade	Gain
Computers	250	270	+20
Wheat	2,500	2,700	+200

Japan

	No trade	Trade	Gain
Computers	120	130	+10
Wheat	600	700	+100

World output: computers $370 \rightarrow 400$, wheat $3,100 \rightarrow 3,400$. Same labor, same technology, bigger pie.

Trade untied consumption from production. It works like a technological improvement.

Where do the gains come from?

Absolute advantage: producing a good using *fewer inputs* than another producer.

- Wheat: 10 hours per ton in the U.S., 25 in Japan. U.S. has the absolute advantage.
- Computers: 100 hours in the U.S., 125 in Japan. U.S. has the absolute advantage.

The U.S. is better at producing *both* goods. So why does it import computers from Japan, and why does Japan gain?

Two measures of cost

Inputs required (absolute advantage)

Hours per unit. The U.S. wins on both.

Opportunity cost

What is given up. The opportunity cost of 1 computer = the wheat that the same hours could have produced.

- U.S.: 100 hours per computer could make $100/10 = 10$ tons of wheat.
- Japan: 125 hours per computer could make $125/25 = 5$ tons of wheat.

A computer is *cheaper* in Japan once cost is measured in wheat given up.

Comparative advantage

Comparative advantage: producing a good at a *lower opportunity cost* than another producer.

Opportunity cost of...	1 computer	1 ton of wheat
United States	10 tons wheat	1/10 computer
Japan	5 tons wheat	1/5 computer
Comparative advantage	Japan	United States

- A country's two opportunity costs are reciprocals, so a country can have an absolute advantage in both goods but never a comparative advantage in both.

The principle of comparative advantage

Principle

Each good should be produced by the party with the lower opportunity cost of producing it.

- Gains from trade come from **differences in opportunity cost**, not from differences in input requirements.
- When each side specializes according to comparative advantage, total output rises and both sides can consume more.
- If opportunity costs are identical, there is no comparative advantage and no gain from trade.

The price of trade

Both gain only if the price of a computer lies between the two opportunity costs: $5 < \text{price} < 10$ tons of wheat per computer.

- Japan's cost is 5 tons; it gains by selling for more than 5.
- The U.S. cost is 10 tons; it gains by buying for less than 10.
- Our trade: 700 tons for 110 computers, $700/110 \approx 6.4$ tons per computer.

Question

What if Japan asks 12 tons per computer, or the U.S. offers only 4?

Argentina and Brazil

Each has 10,000 labor hours per month.

Hours per unit	1 lb coffee	1 bottle wine
Argentina	2	4
Brazil	1	5

Question

Which country has the absolute advantage in coffee? Which has the comparative advantage in wine?

Argentina and Brazil: answers

Opportunity cost of...	1 lb coffee	1 bottle wine
Argentina	$2/4 = 0.5$ bottle	$4/2 = 2$ lb coffee
Brazil	$1/5 = 0.2$ bottle	$5/1 = 5$ lb coffee

- Absolute advantage in coffee: **Brazil** (1 hour vs. 2).
- Comparative advantage in wine: **Argentina** (gives up 2 lb coffee vs. 5).
- Comparative advantage in coffee: Brazil (0.2 bottle vs. 0.5).
- A bottle of wine trades for between 2 and 5 lb of coffee.

Should a star athlete mow his own lawn?

- The athlete mows the lawn in 2 hours. In 2 hours he could film a commercial for \$20,000.
- The teenager next door mows it in 4 hours. In 4 hours he could earn \$40 at a part-time job.
- Absolute advantage in mowing: the athlete.
- Opportunity cost of mowing: \$20,000 for the athlete, \$40 for the teenager.
- Comparative advantage in mowing: **the teenager.**

Any payment between \$40 and \$20,000 makes both better off. This is why people specialize in one job and buy everything else.

Should the U.S. trade with China?

- Same logic as U.S.–Japan: if China's opportunity cost of clothing or electronics is lower, Americans as a whole gain by importing them and shifting labor to their comparative advantage.
- Surveyed economists almost unanimously agree that trade with China makes most Americans better off.

Question

“Some Americans who work in industries that compete with Chinese imports are made worse off.” Agree, disagree, or uncertain?

Economists agree with this too. Trade creates winners and losers; the gains exceed the losses, but compensation is not automatic.

Summary

- Interdependence and trade let everyone consume more, and more variety, than they could produce alone.
- Absolute advantage: fewer inputs. Comparative advantage: lower opportunity cost.
- Gains from trade come from **comparative advantage**. One party can have an absolute advantage in everything; no one can have a comparative advantage in everything.
- Specialize in the good where you have the comparative advantage; trade for the rest.
- The price of trade must lie between the two opportunity costs.
- Applies to individuals and countries alike; it is the basis of the case for free trade.